InCoB/ISCB-APAC 2026

Text to knowledge graph: unlocking science with semantic toolkits

Monday, September 14, 2026

Penang, Malaysia



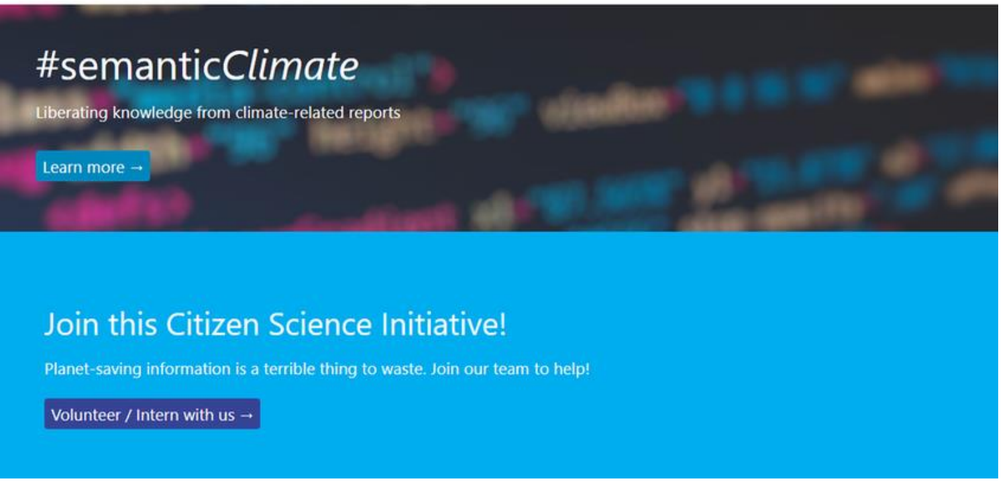
Presented by Renu Kumari

Program Manager, #semanticClimate

Researcher, BRIC-NIPGR, New Delhi, India

AMI





- 6 years old
- Team of interns (Online and offline)
- Work by creating, distributing and using software.

Youth Knowledge Champions

★ It's committed to making public knowledge available in a frictionless form for everybody.



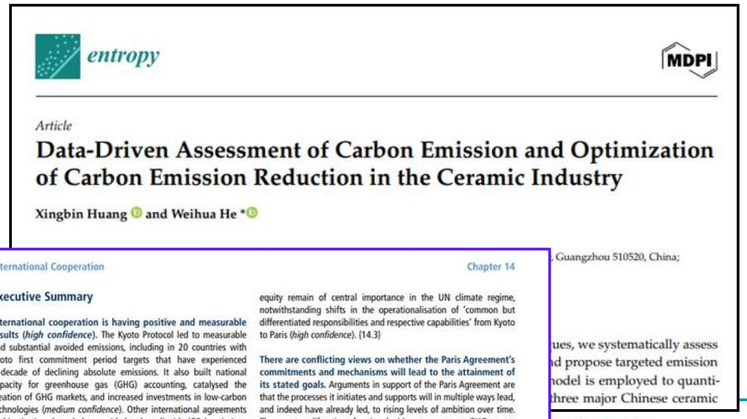





https://semanticclimate.github.io/p/en/

OVERVIEW

Scientific literatures



entropy

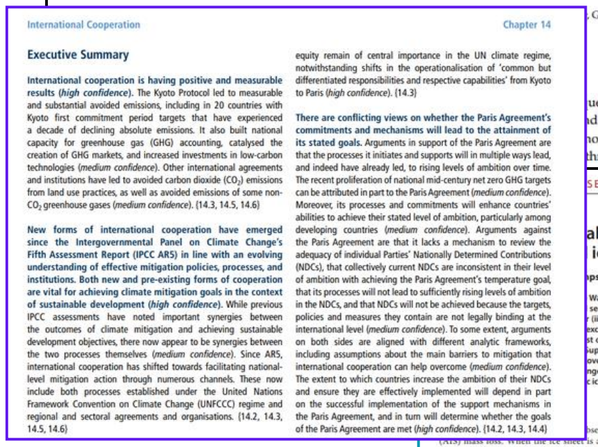
MDPI

Article

Data-Driven Assessment of Carbon Emission and Optimization of Carbon Emission Reduction in the Ceramic Industry

Xingbin Huang and Weihua He *

, Guangzhou 510520, China;



International Cooperation Chapter 14

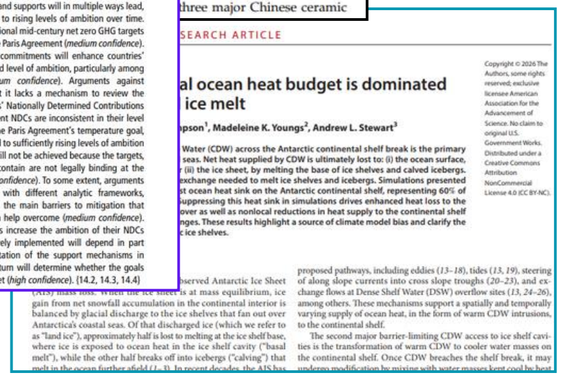
Executive Summary

International cooperation is having positive and measurable results (*high confidence*). The Kyoto Protocol led to measurable and substantial avoided emissions, including in 20 countries with Kyoto first commitment period targets that have experienced a decade of declining absolute emissions. It also built national capacity for greenhouse gas (GHG) accounting, catalysed the creation of GHG markets, and increased investments in low-carbon technologies (*medium confidence*). Other international agreements and institutions have led to avoided carbon dioxide (CO_2) emissions from land use practices, as well as avoided emissions of some non-CO_2 greenhouse gases (*medium confidence*). {14.3, 14.5, 14.6}

New forms of international cooperation have emerged since the Intergovernmental Panel on Climate Change's Fifth Assessment Report (IPCC AR5) in line with an evolving understanding of effective mitigation policies, processes, and institutions. Both new and pre-existing forms of cooperation are vital for achieving climate mitigation goals in the context of sustainable development (*high confidence*). While previous IPCC assessments have noted important synergies between the outcomes of climate mitigation and achieving sustainable development objectives, there now appear to be synergies between the two processes themselves (*medium confidence*). Since AR5, international cooperation has shifted towards facilitating national-level mitigation action through numerous channels. These now include both processes established under the United Nations Framework Convention on Climate Change (UNFCCC) regime and regional and sectoral agreements and organisations. {14.2, 14.3, 14.5, 14.6}

equity remain of central importance in the UN climate regime, notwithstanding shifts in the operationalisation of 'common but differentiated responsibilities and respective capabilities' from Kyoto to Paris (*high confidence*). {14.3}

There are conflicting views on whether the Paris Agreement's commitments and mechanisms will lead to the attainment of its stated goals. Arguments in support of the Paris Agreement are that the processes it initiates and supports will in multiple ways lead, and indeed have already led, to rising levels of ambition over time. The recent proliferation of national mid-century net zero GHG targets can be attributed in part to the Paris Agreement (*medium confidence*). Moreover, its processes and commitments will enhance countries' abilities to achieve their stated level of ambition, particularly among developing countries (*medium confidence*). Arguments against the Paris Agreement are that it lacks a mechanism to review the adequacy of individual Parties' Nationally Determined Contributions (NDCs), that collectively current NDCs are inconsistent in their level of ambition with achieving the Paris Agreement's temperature goal, that its processes will not lead to sufficiently rising levels of ambition in the NDCs, and that NDCs will not be achieved because the targets, policies and measures they contain are not legally binding at the international level (*medium confidence*). To some extent, arguments on both sides are aligned with different analytic frameworks, including assumptions about the main barriers to mitigation that international cooperation can help overcome (*medium confidence*). The extent to which countries increase the ambition of their NDCs and ensure they are effectively implemented will depend in part on the successful implementation of the support mechanisms in the Paris Agreement, and in turn will determine whether the goals of the Paris Agreement are met (*high confidence*). {14.2, 14.3, 14.4}



SEARCH ARTICLE

al ocean heat budget is dominated ice melt

pson[1], Madeleine K. Youngs[2], Andrew L. Stewart[3]

Water (CDW) across the Antarctic continental shelf break is the primary seas. Net heat supplied by CDW is ultimately lost to: (i) the ocean surface, (ii) the ice sheet, by melting the base of ice shelves and calved icebergs. exchange needed to melt ice shelves and icebergs. Simulations presented st ocean heat sink on the Antarctic continental shelf, representing 60% of suppressing this heat sink in simulations drives enhanced heat loss to the over as well as nonlocal reductions in heat supply to the continental shelf nges. These results highlight a source of climate model bias and clarify the c ice shelves.

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bserved Antarctic Ice Sheet

gain from net snowfall accumulation in the continental interior is balanced by glacial discharge to the ice shelves that fan out over Antarctica's coastal seas. Of that discharged ice (which we refer to as "land ice"), approximately half is lost to melting at the ice shelf base, where ice is exposed to ocean heat in the ice shelf cavity ("basal melt"), while the other half breaks off into icebergs ("calving") that

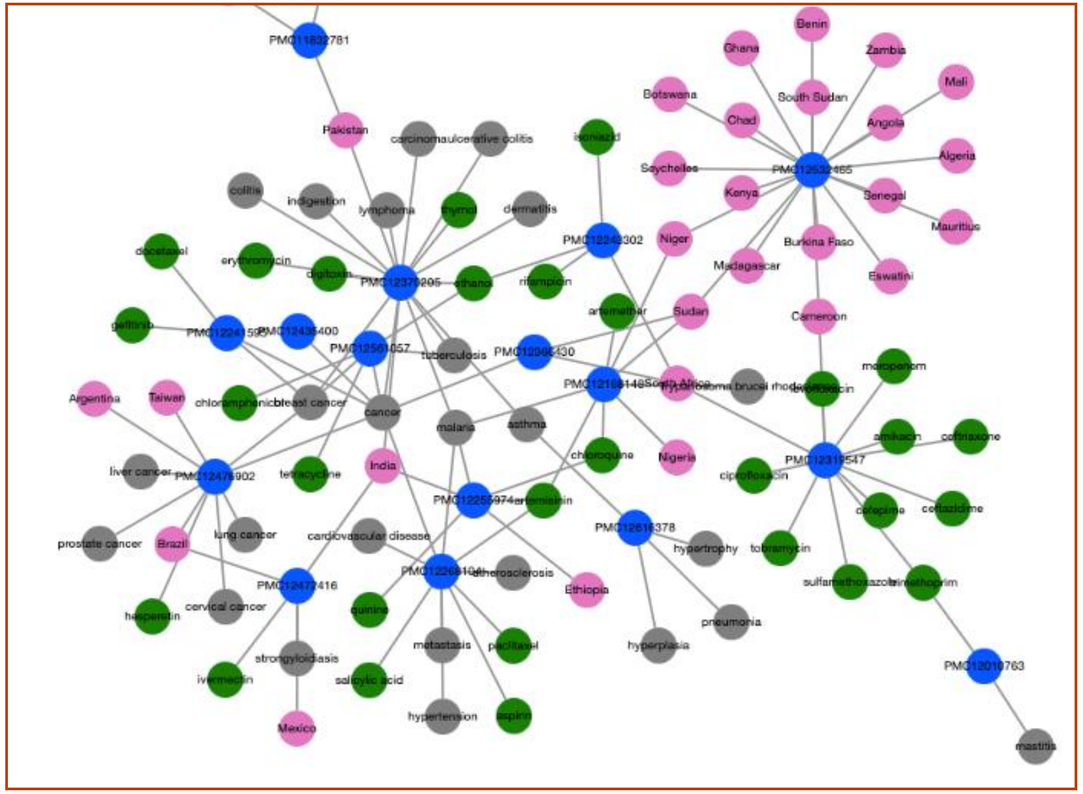
proposed pathways, including eddies (*13*–*18*), tides (*13*, *19*), steering of along slope currents into cross slope troughs (*20*–*23*), and exchange flows at Dense Shelf Water (DSW) overflow sites (*13*, *24*–*26*), among others. These mechanisms support a spatially and temporally varying supply of ocean heat, in the form of warm CDW intrusions, to the continental shelf.

The second major barrier-limiting CDW access to ice shelf cavities is the transformation of warm CDW to cooler water masses on the continental shelf. Once CDW breaches the shelf break, it may

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Transforming information into actionable knowledge

Knowledge Graph



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WHY DO WE NEED THIS?





- Exponential growth of publications
- Difficult to keep up with the latest developments
- Literature exists in various formats: mainly PDFs
- Not machine-readable or structured formats
- Limited Access to the repositories and journals
- No single platform for getting access to all research outputs
- Bulk downloading is often restricted
- Technical Barriers to automate article retrieval for people with no coding

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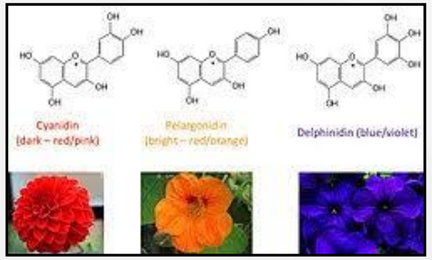
APPLICATIONS



Drugs



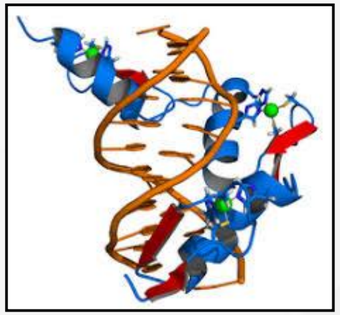
Species



Phytochemicals



Scholarly publications



Genes



Location

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WORKFLOW

Add query term

1 *pygetpapers* — Input

pygetpapers

Open access repository
(ex: Europe PMC)

2 *amilib*

Output

Creates datatables for the corpus

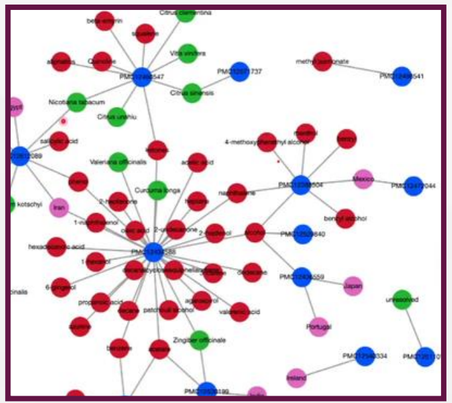
download PDF files

Give papers in XML/HTML format

docanalysis
/NER

3

NER (Species, Drug, Diseases etc.)



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Software

pygetpapers : CORPUS BUILDER

Automated Download of Research Papers from various scientific repositories




developed by: **Ayush Garg and Peter Murray-Rust**

pygetpapers

pygetpapers 1.2.5

```
pip install pygetpapers
```

Latest version

Released: Dec 21, 2022

GitHub repo: https://github.com/petermr/pygetpapers

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Software

ADVANCED FEATURES `pygetpapers`

pygetpapers

:

1 Date Range Queries

Search papers published between dates

--startdate 2023-01-01 --enddate 2023-12-31

2 Term-based Queries

terms.txt with comma-separated terms

drug, natural products, deep learning

Output Formats

1. JSON Metadata
2. CSV Output
3. HTML Tables
4. Datatables
5. Full text PDF

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Software

amilib

Library to support downloading and analysis of documents, mainly from Open Access repositories, from published scholarly articles, or from authoritative sites such as UN IPCC or UNFCCC.



developed by:

Peter Murray-Rust

Professor Emeritus, Cambridge University

and Co-Founder #semanticClimate

amilib 1.1.2

```
pip install amilib==1.1.2
```

Latest version

Released: Nov 24, 2025

Used for semantic indexing of the corpus: Datatable

https://github.com/petermr/amilib

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Software

docanalysis

docanalysis is a Command Line Tool that ingests corpora (CProjects) and carries out text-analysis of documents, including

- **sectioning**
- **NLP/text-mining**
- **dictionary generation**



Used for named entity recognition (NER)

developed by:





Shweata N Hegde and

Peter Murray-Rust

https://github.com/petermr/docanalysis

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Platform to run the softwares/ tool

Google Colab Notebook

Importance of Colab Notebook



- Open Jupyter E-Notebook environment
- No pain with setups, versions
- Human Machine friendly
- Supports interactive programming
- Easy learn and explore new tools

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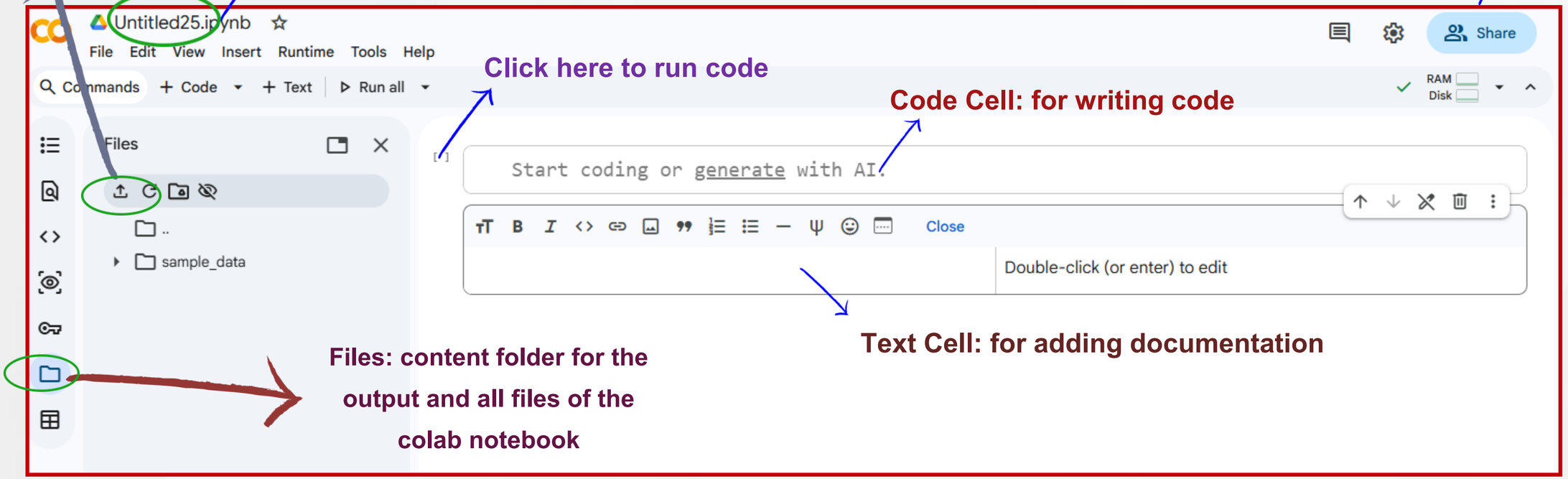
GOOGLE COLAB (COLLABORATORY)

free, cloud-based platform to write and execute python-based codes

Upload files/folders

Title of the colab notebook

share the notebook



Click here to run code

Code Cell: for writing code

Files: content folder for the output and all files of the colab notebook

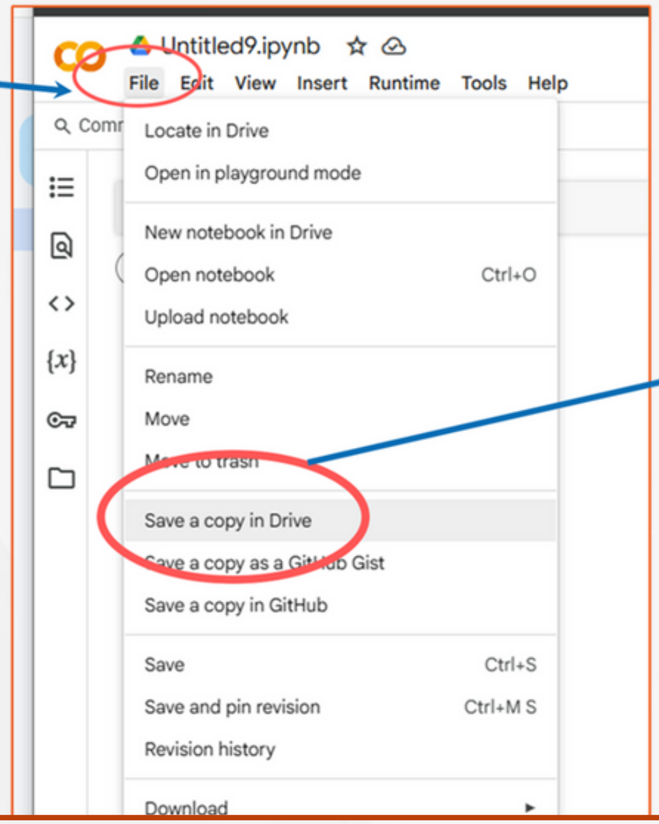
Text Cell: for adding documentation

Need Google account to get started!

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GOOGLE COLAB (COLLABORATORY)

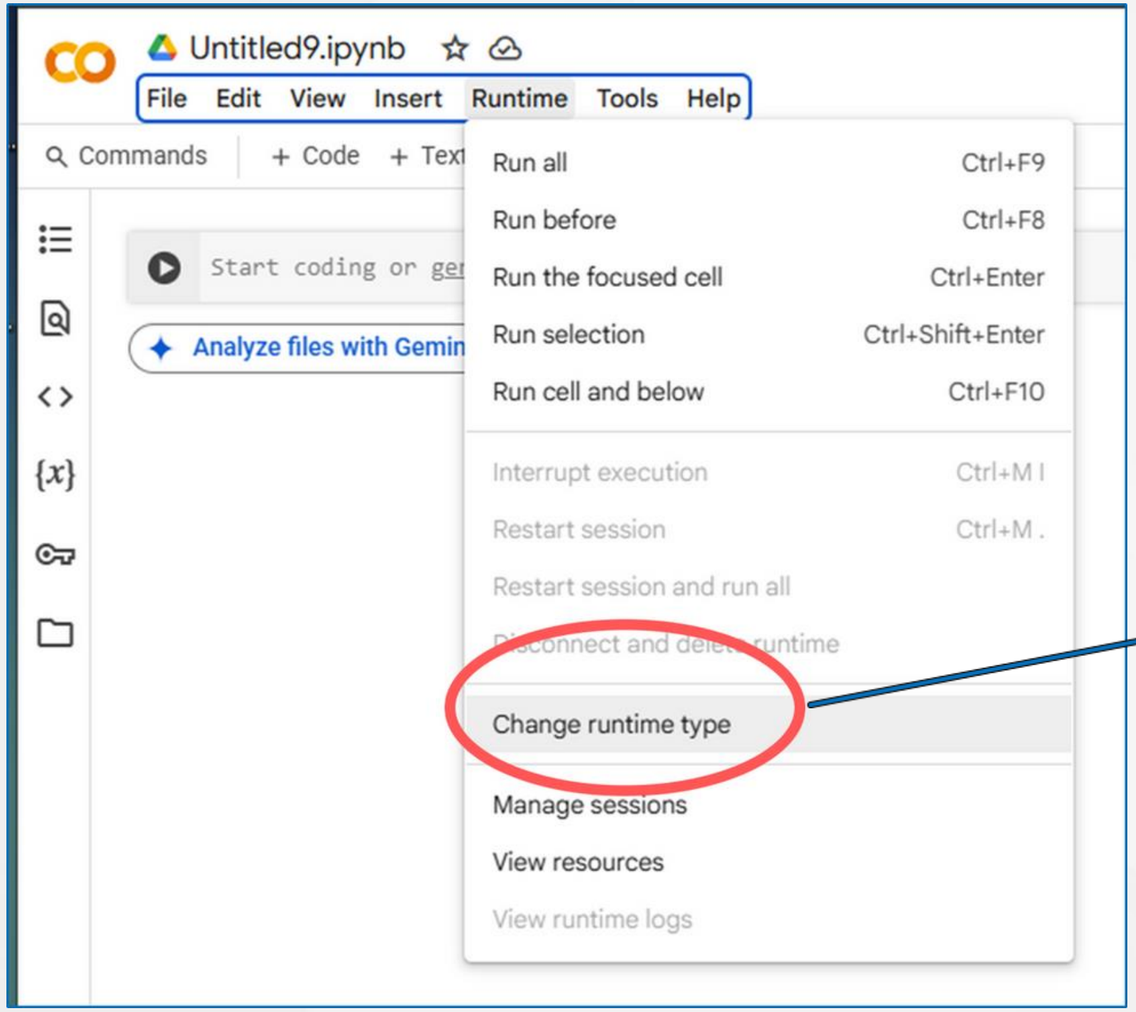
Click to File



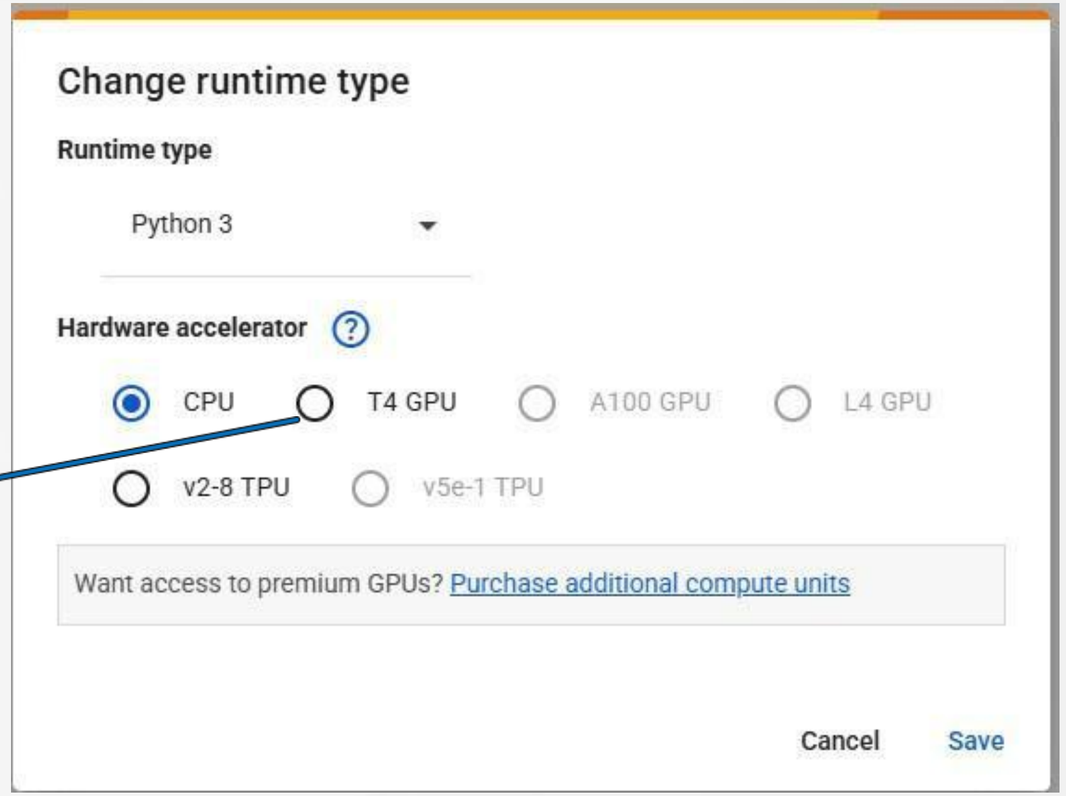
Save a copy in Drive

Before using this Colab, Save a copy to your own Google Drive:
Click on “File” > “Save a copy in Drive”

CHANGE RUNTIME FOR MORE MEMORY REQUIREMENT



T4 GPU



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Querying *pygetpapers* in colab notebook

Download the 50 Most Recent Articles for a Search Query Within a Specific Time Frame.

time taken: 1-3 minute

```
[4]
✓ 1m
!pygetpapers --query '"drug discovery" AND "medicinal plants"' --xml --limit 50 --startdate "2010-01-01" --enddate "2025-10-31" --output drug_discovery --save_query

INFO: Total Hits are 6789
50it [00:00, 127177.20it/s]
INFO: Saving XML files to /content/drug_discovery/*/fulltext.xml
100% 50/50 [01:06<00:00,  1.33s/it]
```

Search query

XML

Number of papers

!pygetpapers --query '"drug discovery" AND "medicinal plants"' --xml --limit 50 --startdate "2010-01-01" --enddate "2025-10-31" --output drug_discovery --save_query

timeframe

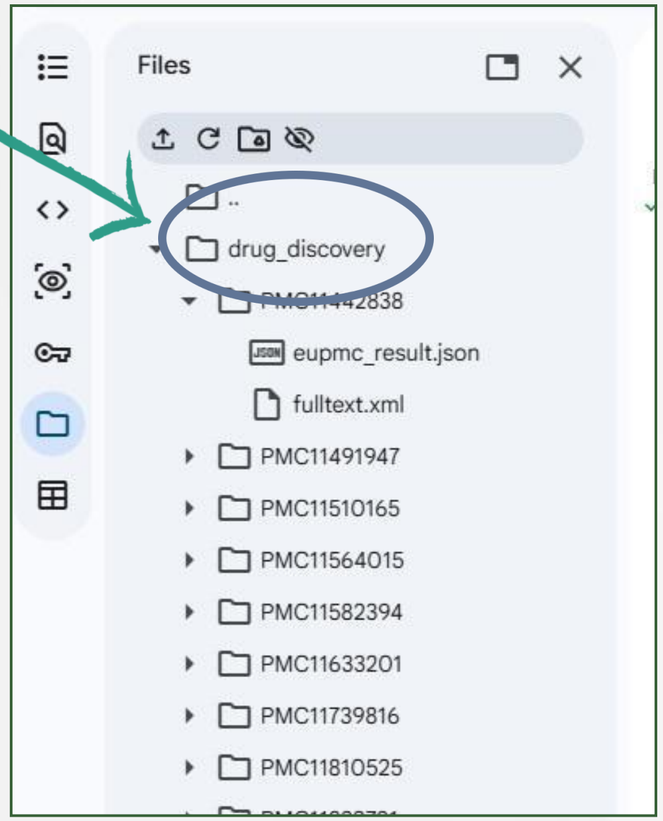
Output folder

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Transforming information into actionable knowledge

Results : semantic corpus for the query

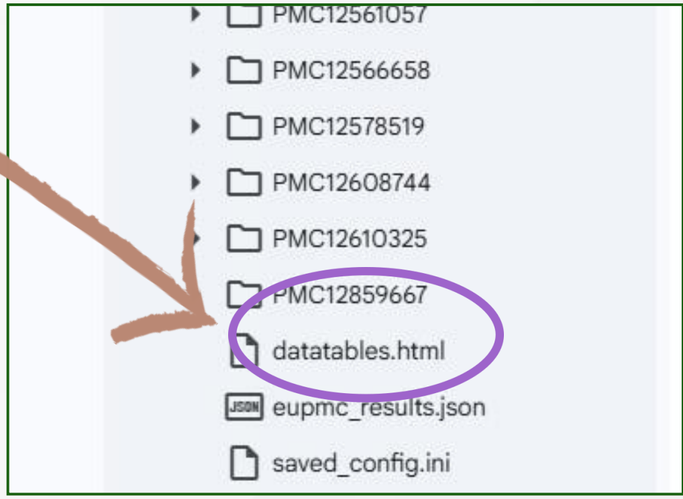
Output :

drug_discovery



Summary table

(Semantic Index)



Query:

"drug discovery" AND "medicinal plants"

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Semantic Index of the Corpus

Datatable

- PMCID (for Europe PMC)
- DOI,
- Title,
- Authors,
- Journal name etc.....

10 entries per page

Search:

query: "drug discovery" AND "medicinal plants"; start: 2010-01-01; end: 2025-10-31

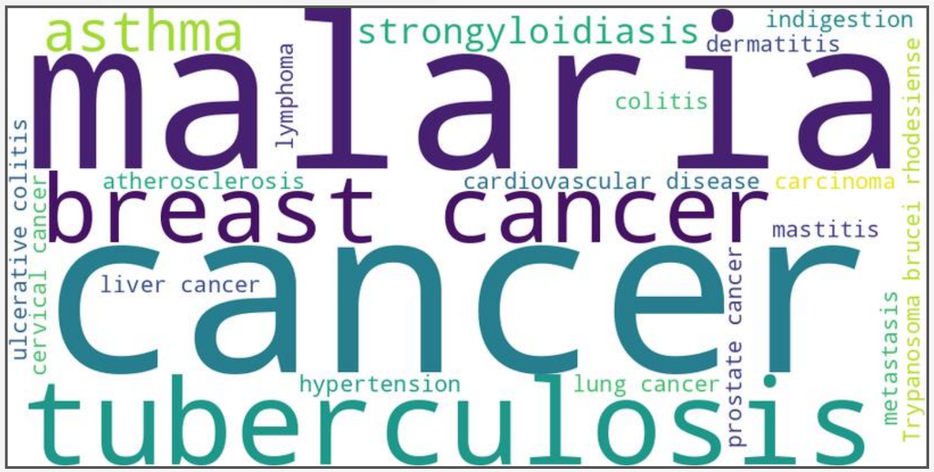
pmcid	doi	title	authorString	journalInfo.journal.title	pubYear	abstractText
PMC11442838	10.3389/fphar.2024.1488930	Editorial: Global excellence in ethnopharmacology: North and South America.	• Cristians S • Mata R.	Frontiers in pharmacology	2024	
PMC11491947	10.1016/j.jsps.2024.102185	The Dawn till Dusk of phytopharmaceuticals.	• Tiwari R • Mishra S • Chauhan A • Gulati P • Dhobi M.	Saudi pharmaceutical journal : SPJ : the official publication of the Saudi Pharmaceutical Society	2024	Herbal products and their formulations large market at the global level. A signi portion of the worldwide population rel herbal treatment. Their apparent non-to cost-effect
PMC11510165	10.3390/ph17101307	Plant Metabolomics: The Future of Anticancer Drug Discovery.	• Dabbousy R • Rima M • Roufayel R • ... • Sabatier JM • Fajloun Z.	Pharmaceuticals (Basel, Switzerland)	2024	Drug development from medicinal plan constitutes an important strategy for fin natural anticancer therapies. While seve secondary metabolites with potential an activities have be

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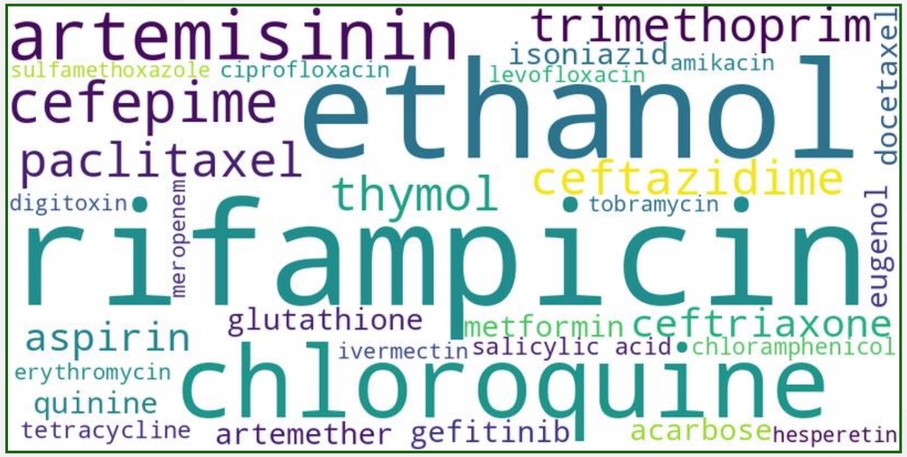
Result: Extracted entities from the literature



Countries (Location)



Diseases



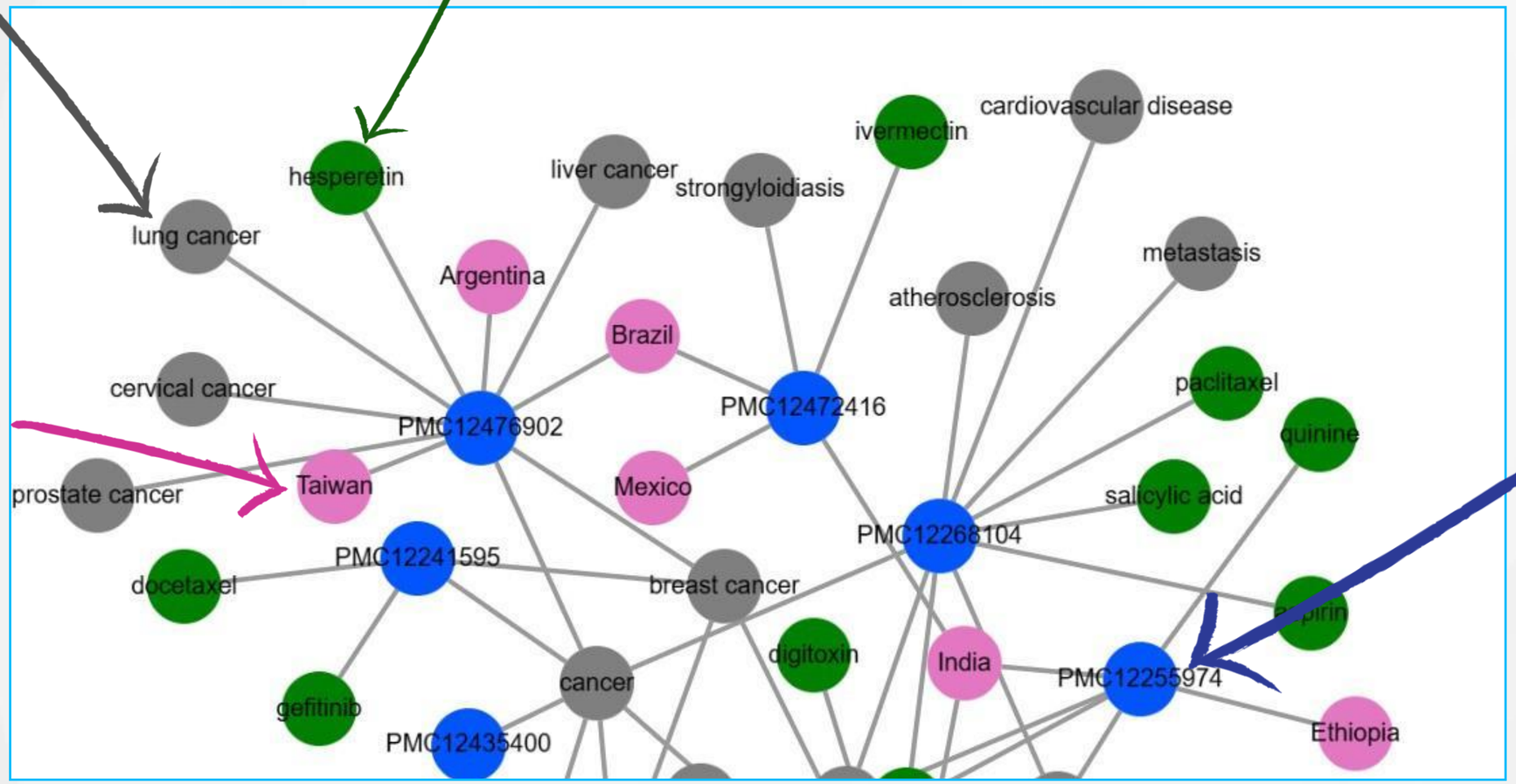
Drugs

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Result: Knowledge Graph

Disease

Drugs/ Chemical compounds



Country (location)

scholarly publication

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Resources



E-mail semanticclimate@gmail.com

Website https://semanticclimate.github.io/p/en/

GitHub https://github.com/semanticClimate

GitHub Repositories for softwares

pygetpapers: https://github.com/petermr/pygetpapers

txt2phrases: https://github.com/semanticClimate/txt2phrases

semantic_corpus: https://github.com/semanticClimate/semantic_corpus

encyclopedia https://github.com/semanticClimate/encyclopedia

amilib : https://github.com/petermr/amilib

CABot : https://github.com/semanticClimate/semantic_RAG

ClimateInsight: https://github.com/semanticClimate/ClimateInsight

Climate Academy Chatbot (CABot)

https://cabot-semanticclimate.org

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बीआरआईसी
बायोटेक्नोलॉजी विभाग का संगठन
BRIC
a DBT Organization

COMMITTEE ON DATA
CODATA
INTERNATIONAL SCIENCE COUNCIL

InCoB/ISCB-APAC 2026

ISCB
INTERNATIONAL SOCIETY FOR COMPUTATIONAL BIOLOGY

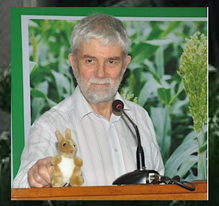
THANK YOU ALL FOR JOINING US



Dr. Gitanjali Yadav



Prof. Peter Murray-Rust



Priti



Mandeep



Simon Worthington



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Hands-on Session

Link to Colab Notebook

https://colab.research.google.com/drive/15NaDyXl1uzQcBxWZg0r-jAQGKLjlWrs4?usp=sharing

short url to Colab Notebook

https://bit.ly/ALR-semantic